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Efficiency of digital and non-digital educational games: A comparative meta-analysis and a meta-thematic analysis

Tarik Talan^a, Yunus Doğan^b , and Veli Batdı^a 

^aKilis 7 Aralık University, Kilis, Turkey; ^bFirat University, Elazığ, Turkey

ABSTRACT

The present article attempts to reinterpret the findings of most recent studies investigating effect of using games for teaching purposes. A methodological approach combining a meta-analysis of quantitative data with qualitative ones was adopted in order to present the broadest picture of the current research on educational use of games. To this end, we conducted a meta-analysis of 180 effect size comparisons out of 154 empirical studies on the effect of both digital and non-digital games on academic achievement conducted during the period from 2004 to 2019 in order to determine the overall effect size of using games for teaching various subjects. The overall sample size of the studies included a total number of 12800 participants. Some moderator analyses were also carried out to determine the exact efficiency of educational games in terms of student levels, durations of implementation of game activities, school subjects in which games were used, class sizes, kinds of games and achievement tests used. The findings suggest that educational games have a positive effect on academic achievement and this effect is at a medium level ($g=0.695$). The highest effect sizes were observed in foreign language courses ($g=0.87$), small (less than 50) class sizes ($g=0.87$), and in non-digital games ($g=0.90$). Moreover, we conducted a meta-thematic analysis based on document analysis of qualitative studies in order to further consolidate the findings of the meta-analysis. The meta-thematic dimension of our study reveals cognitive contributions as well as drawbacks of game-based teaching, and provides suggestions for conducting educational games in a better way.

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Game; educational game; digital game; non-digital game; meta-analysis; meta-thematic analysis

Introduction

Curriculum developers, instructional designers and practitioners of any specific area of human learning highlight the use of various instructional techniques which are tailored according to learners' characteristics and needs, and that could facilitate fulfillment of pre-specified learning objectives. Whether the learning setting is a traditional classroom or a virtual environment, one of these instructional techniques exactly pinpointed in contemporary educational theory is utilization of educational games to support the learning-teaching process. Since the inception of constructivist theory centering on individual as an active agent building knowledge and meaning based on personal experiences and hypotheses within social context, and the social cognitive theory emphasizing the social influence on occurrence of learning (Schunk, 2008), games as pedagogical devices have been on the agenda of both educational research and practice as they are thought to provide the required social context in which real-life experiences are simulated with

reciprocal interaction, observation and modeling (Hense & Mandl, 2012). Thus from an educational psychology perspective, educational games have been regarded to offer suitable premises for individual's physical, cognitive and affective development throughout the growth periods of childhood, puberty and young adulthood (Alexiou & Schippers, 2018; Elkonin, 2005; Gee, 2007; Granic, Lobel, & Engels, 2014; Nisbet & Williams, 2009; Shapiro, 2014; Woolfolk, 2018).

Games which were traditionally already around as a tool of socialization and cooperatively learning among children have then begun to be incorporated into learning environments more often, and along with the technological advancements and emergence of new learning settings with online applications and virtual worlds (Dagleish & Launrenson, 2019; Vallance & Martin, 2012), new merits in terms of learning accomplishment and its components like, inter alia, motivation, engagement, retention, critical thinking, etc. are still being reported in favor of game activities in spite of the change in their structure and administration. This being the case, a synthesis of both quantitative and qualitative relevant research clarifying a broader picture of the efficiency of games for instructional purposes in various areas of learning comes forth as a crucial endeavor in the theoretical dimension.

Literature review

Reframing games within educational context

According to Salen and Zimmerman (2004) play and games have historically been studied in a myriad of ways, and any proposed definition of games would depend on the field in which it is meant to be used. As a characteristic unique to humans, the definition of a game, whether including use of any technology or not, is cited in the broadest sense as “an activity governed by precise rules that involve varying degrees of chance or luck, and one or more players who cooperate or compete (with oneself, the game, one another or a computer) through the use of knowledge or skill in an attempt to reach a specified goal” (Jaffe, 2007). Although they are at the same age with humanity as instruments of discovery and interaction, the inclusion of games into modern teaching methodologies are said to have been occurred with the influence of such scholars as John Dewey (Gros, 2003, Schultz, 2017), Jean Piaget (Linaza, 1984) and Lev Vygotsky (Bodrova & Leong, 2015). While Dewey gave games as an example of ‘how societies reproduce themselves and structure the interaction patterns of their citizens’ and thought that games bring attitude changes to children and make them more critical thus they can be given tasks and assume responsibilities (Schultz, 2017), Piaget regarded games as the last demonstration of children's playfulness, and he considered of a close interaction between intellectual growth and game activity (Linaza, 1984). And we understand from the Vygotskian view that well-formed games can be a source of the zone of proximal development in childhood (Bodrova & Leong, 2015), as they expose a child to a learning condition which can merely be situated within his/her area of close development (Annetta, 2008). In fact, there is a sound literature accumulated on philosophical and psychological foundations of educational use of games, thus it would be going beyond the scope of this study to further iterate what psychological and educational theories propose about the place of games in human learning; and thus it would be more appropriate here to dwell upon what the empirical evidence puts forth over learning based on games.

Games, which ensure a link between theoretical learning and its applications, have constituted a significant part of the modern educational strategies employed by teachers at various levels of schooling ranging from kindergartens to universities since first scientific evidence shed light on their usefulness in learning activities. As Jaffe (2007) reports, games began to be used to meet the goal of promoting the experience-based learning which became prominent in the 1950s and 1960s as the theoretical focus of learning changed from the instructor to the student. In a different fashion, Egenfeldt-Nielsen (2007) attributes it implicitly to a social transformation and states

that traditional teaching methods were not regarded as capable of preparing students for the increased complexity in society and the learning skills necessary in real-life, thus the complicated nature of games which potentially prepare students for the difficulties in modern society was seen as a premise of educational use of games. With the ever-increasing digitalization of learning environments, games which were once utilized only as supportive instructional techniques in addition to conventional classroom teaching now turns out to be a whole approach to learning as “game-based learning (GBL),” which is defined as the utilization of game-based technology to deliver, support and improve teaching, learning, assessment and evaluation (Vlachopoulos & Makri, 2017).

It is discernable from the relevant literature that there is not a clear unanimity over definition of the GBL, but a tendency to confine it with the use of computer or video games in instructional implementations (gamification) prevails (Groff, McCall, Darvasi, & Gilbert, 2016; Kapp, 2012; Plass, Homer, & Kinzer, 2015). On the other hand, Plass et al. (2015) point to a distinction between the GBL and gamification by emphasizing that the “game aspect” in the GBL is not always a digital one, but any type of game play with defined learning outcomes, while gamification “involves the use of game elements, such as incentive systems, to motivate players to engage in a task they otherwise would not find attractive (p.259).” In accordance with the nature of our study and as it is used by some other researchers as well (Habgood & Ainsworth, 2011), we tend to use the term GBL in its broadest sense to include any instructional activity based on the use of games, whether they are conducted traditionally (non-digital classroom games specific to teaching of school subjects) or via digital technology (video games, virtual worlds, simulation games, etc.).

Effects of games on learning components

It is evident that there is ample research conducted on the educational uses of games and its effectiveness in terms of learner achievement and cognitive and affective aspects of learning contributing to achievement. As to the cognitive gains obtained with the use of games in learning environments in the general sense, a significant increase is reported in such salient cognitive qualities of learners as problem-solving skills (Adachi & Willoughby, 2013; Granic et al., 2014; Hwang, Wu, & Chen, 2012; Justice & Ritzhaupt, 2015; Ritzhaupt, Gunter, & Jones, 2010; Spires, Rowe, Mott, & Lester, 2011; Yang, 2012), critical thinking and student creativity (Cicchino, 2015; Hallajian, 2016; Naeini & Masood, 2012), memory (Motabarzadeh & Musavi, 2015; Tavarez, 2012) knowledge retention and recall (Abdullah, Abu Bakar, Ali, Faye, & Hasan, 2012; Alici, 2016; Babaandaç, 2013; Bayram, 2015; Selvi & Çoşan, 2018; Shabaneh & Farrah, 2019), cognitive abilities like analogy, processing speed and deductive reasoning (Hisam et al., 2018), perceptual attention and mental rotation skills (Alexiou & Schippers, 2018; Mayer, 2019), and spatial skills which are purported to predict achievement in STEM areas (Granic et al., 2014). In their comprehensive meta-analysis of 248 studies covering the period of 1986-2003, Vogel et al. (2006) concluded that the utilization of interactive games enabled to obtain higher cognitive gains. Furthermore, there are also diverse elements in games which may appeal to different individual learning styles (De Byl & Brand, 2011; Sugar & Sugar, 2002).

There is also a current interest in researching the positive impacts of the GBL on the dimensions of affect in learning like motivation, attitudes, engagement, self-efficacy, self-esteem, social recognition, and anxiety (Aksoy, 2014; Alexiou & Schippers, 2018; Annetta, 2008; Chow, Woodford, & Maes, 2011; Ebrahimzadeh & Alavi, 2017; Hense & Mandl, 2012; Ingram & Cangemi, 2019; Jaffe, 2007; Ke, Xie, & Xie, 2016; Ritzhaupt, Higgins, & Allred, 2011; Şahin, 2016; Tiede & Grafe, 2018; Yazıcıoğlu, 2017), and a more recent trend toward identifying what the GBL provides in terms of the 21st century skills such as innovative thinking, collaboration, information and technology literacy, productivity, etc. (Annetta, 2008; Bakan & Bakan, 2018; Qian & Clark, 2016).

Barriers to adoption of games for educational purposes: Criticisms and counter-evidence

The GBL is certainly not devoid of criticisms raised against itself, particularly in terms of digital games; in that some oppositions and barriers to incorporating computer games and simulations in the teaching-learning process are reported to be voiced by teachers and parents (Baek, 2008; Byron, 2008; Furdu, Tomozei, & Kose, 2017; Gros, 2007; Justice & Ritzhaupt, 2015; Kenny & Gunter, 2011; Plowman, McPake, & Stephen, 2010; Ritzhaupt et al., 2010). However, as it is called for by Granic et al. (2014), a more balanced perspective should be taken while making evaluations over the possible negative effects of these games by also considering their benefits. A strong concern for the possible emergence of addictive behavior (Griffiths, 2004, 2005), aggression (Fletcher & Tobias, 2006) and extreme competition with the use of some digital games is first and foremost said to inhibit teachers from utilizing them in the classroom (Baek, 2008). Yet, a more recent longitudinal study by Kühn et al. (2019) examining the long-term effects of video games nullifies the worries about subsequent possibility of aggressive behavior by reporting no significant results. Another worrying presumption that has necessarily preoccupied parents and teachers against the use of digital games for instruction is association of gaming with poor academic achievement (Fletcher & Tobias, 2006; Kennard, 2018). Nevertheless, there are some empirical studies showing no negative effect of playing digital games on school grades (Drummond & Sauer, 2014) or even suggesting positive impacts on achievement in some school subjects (Posso, 2016; Tannahill, Tissington, & Senior, 2012).

The particular focus on overusing such motivators as points and badges and moving away from its educational purpose and turning a game into a mechanical behavior is said to lead to the loss of student interest after a while (Furdu et al., 2017) and a decrease in the intrinsic significance of the game essential elements and objectives (Silpasuwanchai & Ren, 2018). In this sense, Alexiou and Schippers (2018) suggest some theoretical establishments that are supposed to enhance and sustain players' cognitive and emotional engagement: setting imminent goals rather than distant ones, inclusion of indefiniteness into reward systems of games, providing on-time and frequent feedback, and incorporation of some hedonic game elements like esthetically luring audiovisual components identifying players with game characters. Another barrier to utilizing digital games is reported to be whether the teaching platforms would be conducive to achieving targeted learning objectives, and accordingly Moizer, Lean, Towler, and Abbey (2009) emphasize the critical significance of assessment of convenience of any certain technique to be used in the implementation of such games. In this respect, Yang (2012) concisely cites what digital games design provide for learning as: incorporation of practice and drills ensuring learners an easier retention of information, hypothesis testing and active learning, and allowing learners to have chances to assess themselves via scoring and level skipping.

Moreover, arrangement of class periods as possible as to be compatible with engagement for a longer time in more complex games also hinders classroom implementation of digital games (Gros, 2007; Rice, 2007). On the other hand, Kenny and Gunter (2011) attribute the repudiation of digital games by teachers to the nonexistence of proper instructional design principles, which, as they believe, culminate in engagement and entertainment but non-accomplishment of the intended content. However, there are studies citing teachers' contentment due to successful implementation of digital games in learning settings (Groff, Howells, & Cranmer, 2010; Gros, 2007; VeraQuest, Inc., 2012).

Ritzhaupt et al. (2010) summarize some noteworthy barriers to adoption of such games for instructional purposes as obligation of alignment with the curriculum, negative attitudes toward games, lack of background and financial resources and evidence to back up their utilization and shortness of time for their integration into the teaching process. In line with the points raised here, we decided to make an overall assessment of the drawbacks of using games in learning environments based on the synthesis of some qualitative studies investigating practitioners and learners' perceptions in this sense.

Previous studies and the contribution of the present study

Although it seems that taking a short glance at the relevant literature presents us a myriad of studies on pros and cons of games, the research on educational use of games is still not fully-fledged in the sense of synthesizing their practical aspect; and it is observed that most of the above-mentioned studies and those existing in the literature but not cited here are on the use of digital games played with the assistance of computers or other electronic tools (Ritzhaupt, Poling, Frey, & Johnson, 2014); and the research on the effect of non-digital games or traditional classroom games is either old-fashioned or limited. This fact is articulated also in the literature on use of educational games; and it is noted that there are reviews examining various significant aspects of games promoting learning but are quite dated (Tomlinson & Masuhara, 2009), which is said to be a sign of the decline of research into educational games in favor of research on educational potential of computer games (Egenfeldt-Nielsen, 2007). In other words, the role of non-digital games in pedagogy is often taken for granted, and the empirical research in this respect has been overlooked, and has resulted in an increasingly siloed literature accumulating merely on behalf of digital games.

Moreover, what is also understood from the relevant literature is that while there is a growing number of research studies carried out to investigate and synthesize the experimental evidences of gaming for learning (Bakan & Bakan, 2018; Chen, Tseng, & Hsiao, 2018; Chian-Wen, 2014; Clark, Tanner-Smith, & Killingsworth, 2016; Egenfeldt-Nielsen, 2007; Fletcher & Tobias, 2006; Ke et al., 2016; Liao, Chang, & Chang, 2010; Perrotta, Featherstone, Aston, & Houghton, 2013; Qian & Clark, 2016; Randel, Morris, Wetzel, & Whitehill, 1992; Tokac, Novak, & Thompson, 2019; Vandercruysse, Vandewaetere, & Clarebout, 2012; Yildirim & Şen, 2019), they are largely either systematic reviews or quantitative meta-analyses specific to one school-subject, and are mostly showing the case of computer games and simulations. As an earlier research synthesis, Randel et al. (1992) reviewed the literature on computer games and non-computerized simulations compared to traditional instruction with 68 studies from 1963 to 1991, 56% of which revealed no difference. As a follow-up review of literature updating that of Randel et al. (1992), Fletcher and Tobias (2006) surveyed the empirical research on computer games and simulations between the years 1992-2005 by making theoretical evaluations under such themes as transfer to real life tasks, effects on cognitive processes, design issues, and attitudinal, motivational and affective outcomes. Clark et al. (2016) carried out a meta-analysis and systematic review of 69 study samples ($k = 6,868$) of digital games published between 2000 and 2012 with a synthesis of media comparisons as well as value-added comparisons, and revealed the significance of design beyond medium while assessing the effect of digital games on learning. As a more recent meta-analysis, Yildirim and Şen (2019) reviewed 45 experimental studies ($k = 3,487$) on effect of gamification, which they define as using game design in non-game contexts, on academic achievement, concluding that gamification has added a 7.2% positive value to achievement.

The need for conducting more meta-analyses and systematic reviews on specifically digital gaming for instruction is expressed also in the literature by drawing attention to their scarcity (Fletcher & Tobias, 2006; Mayer, 2019; Ritzhaupt et al., 2014). Thus, this makes our study a significant contribution to the literature, as we conducted a comprehensive meta-analysis of both quantitative and qualitative recent studies in which the effectiveness of digital and non-digital instructional games are questioned in terms of various areas of learning. In other words, the need to re-interpret the cumulative findings of most current research on educational use of games and to pave the way for new studies in this respect led us to design this two-dimensional study with a meta-analysis and a meta-thematic analysis, which is not encountered in the relevant literature. Therefore, it is hoped that this study will provide a unique and integrating framework for understanding the role of games in academic achievement.

Adding a meta-thematic dimension to the study enabled us to recombine common views obtained from qualitative studies dwelling on educational games. What's more, we also conducted

some moderator analyses for the effect of educational games on academic achievement and tried to find out whether academic achievement in question changes in terms of grade levels, duration of game activities, school subjects in which games were conducted, sample sizes and kinds of game activities. The fact that the study pursues a more thorough investigation of the relevant subject-matter in this sense shows its distinction among similar studies.

Moderator variables

Within the meta-analytic investigation of our study, we decided to see whether the overall effect size of educational use of games is moderated by some characteristics of the studies that were included in the meta-analysis. Thus, it would be wise to present a concise rationale for the selection of these moderating variables. Some researchers (Justice & Ritzhaupt, 2015; Ritzhaupt et al., 2014; Kenny & Gunter, 2011) accept unanimously the significance of identifying certain characteristics that may affect instructional design of gaming interventions, and thus contributing to the potential adoption of games by their users within the educational context. A synthesis of research on specifically digital gaming in education by Ritzhaupt et al. (2014) reveals deficiency of completeness in terms of reporting crucial information such as intervention duration, gaming settings and number of players in a game. In this regard, while coding the data of the studies for meta-analysis, some common characteristics like teaching levels, kinds of gaming intervention, duration of game activities, subject areas of teaching, number of students engaged in gaming (class sizes), and the types of testing used in the studies to assess academic achievement drew more attention. In this sense, Randel et al. (1992) believe that an evaluation of research on effectiveness of games with regard to subject areas is valuable since there are significant differences in content and approach adopted in these areas.

Hence, conducting moderator analyses on these characteristics in question was thought to specify the relevant effect size in a more detailed way. It was suggested in this sense that the interpretations regarding moderating variables should be made with caution even if their sample sizes are small (Lesley, Clarke, & Tierney, 1995). Answering such questions as how long game activities should be conducted (Clark et al., 2016; Svingby & Nilsson, 2011), what student-levels are more suitable for teaching through games (Yıldırım & Şen, 2019), in what classes they can be better applied, with what size of class (number of learners) it could yield more fruitful results to conduct game activities and whether the assessment of academic achievement could show higher effect size with a standard test or a test made by the practitioner will provide a framework of reference for both researchers and practitioners. In this respect, a study by Bakker et al. (2019) reveals that researcher-made measures produce larger effect sizes. And this distinguishes our study from other meta-analyses existing in the literature in terms of scope and content as well, despite showing similarities as regards its methodological process to some extent. Meanwhile, a scan of the studies shows us that digital games category consists of computer games and simulations specifically designed for teaching a certain subject-matter; while the studies within the non-digital games context utilized a variety of physical board or card games. Out of these two, an emphasis is given more on digital games with the judgment that learners of this 'digital natives' age are impressed more by digital games (Gros, 2007; De Castell & Jenson, 2007).

The moderator variables of the present study were thus identified as follow:

- Students' Levels: refer to educational stages categorized as Preschool, Primary School, Middle School, High School and University.
- Kinds of Games: refer to digital and non-digital games specifically designed for teaching of a particular subject matter.

- **Duration of Implementation:** refers to the periods during which game interventions were implemented. These periods of time were categorized as 1-4, 5-6, 7-8, 9-18 weeks, Unspecified and Other.
- **School Subjects:** refer to subject areas of teaching in which educational games were used (Science, Social Sciences, Math, Computer, Foreign Languages).
- **Class Sizes:** refer to the number of students engaged in game interventions, and were grouped as Large ($n = 100$ and over), Medium ($n = 50-99$) and Small ($n = \text{less than } 50$).
- **Types of Tests:** refer to the tests used for measuring academic achievement as a result of game implementations, and were classified as Researcher-made Tests (RT), Standardized Tests (ST) and Unspecified (U).

Objectives of the study

In the light of the above-mentioned points, this study aimed to analyze and reinterpret the experimental studies conducted independently of each other to research the implementation of games for educational purposes. The primary purpose for conducting the study was to examine the general impact of both digital and non-digital educational games on academic achievement. The academic achievement refers to the experimental and control groups' post-test scores. Within this framework, we adopted a methodological process involving the simultaneous use of qualitative as well as quantitative methods. To this end, the following research questions were raised and addressed within the study:

1. What is the overall effect size of educational games on learners' academic achievement?
2. Are any effects of educational interventions of games moderated by such certain characteristics of the implementation as *students' levels*, *duration of implementation*, *school subjects*, *class sizes*, and *kinds of games*?
3. What are participants' perceptions about the use of educational games within the context of thematic examination based on document analysis?

Methods and results

As the study is made up of a mixed methods character of research design, the methodological process and results are dealt with separately in both quantitative and qualitative sections. The first section expounds the details of the meta-analytic investigation utilized to combine the results of the quantitative studies on educational games, and the second section explains the dimensions of meta-thematic analysis of the qualitatively conducted studies on the same subject.

Meta-analytic methodology

In the quantitative dimension of the study, in order to identify the effect of the educational games on the academic achievement, the relevant studies in the literature were examined via the method of meta-analysis, representing the process of combination and reevaluation of the results of independent studies carried out on a certain topic (Cohen, Manion, & Morrison, 2002; Littell, Corcoran, & Pillai, 2008). Furthermore, in order to determine the effect sizes of them, the moderator analyses of some variables thought to affect the results of those studies on educational use of games were also conducted.

Data collection

The targeted studies on educational games were searched in the databases of Web of Science, ERIC (EBSCO), Wiley Online Library Full Collection, Science Direct, Taylor & Francis Online, Scopus (A&I), Springer LINK, ProQuest Dissertation & Thesis Global and Turkish Council of Higher Education Thesis Center. In order to reach the relevant research studies, the concepts of “educational games,” “game-based learning,” “educational game,” “game and achievement” were used in the abstracts, index and title search parts of the databases. Our research comprises the studies investigating the variable of academic achievement during the period of 2004-2019.

The academic achievement was determined in the studies included in the meta-analysis via mostly achievement tests prepared and validated by the researchers themselves or with the use of some standardized tests unique to the targeted subject-matter. In terms of research design, the studies were organized mostly with the quasi-experimental design, though some as randomized controlled trials. Within the studies, the trial groups instructed with game activities are expected to differ from the control group without gaming intervention with regard to degree of learning achievement. To this end, both groups are graded in terms of level of learning achievement by conducting a pretest before the gaming intervention and a post-test after the gaming intervention. The marks the students get at the end of the gaming intervention determine their level of achievement. Thus, we tried to reach the studies conducting pretest-posttest designs and calculating achievement scores as inclusion criteria.

Inclusion and exclusion criteria

Studies which did not comply with the inclusion criteria were excluded from the research. As for the other inclusion criteria: (1) the research was limited only with the last fifteen years, as there are numerous studies carried out on the integration of games into education over the last years. (2) The English and non-English (Turkish) research studies published in international peer-reviewed journals and unpublished dissertations and theses conducted during the period 2004-2019 were searched for inclusion. The inclusion of non-English studies and unpublished dissertations/theses into the meta-analysis may be regarded as a sign of lower methodological criteria thresholds (Morrison et al., 2012); however a literature review or systematic review of earlier studies is in fact a process of congregating all studies as possible as a researcher could do. No major differences are reported in meta-analyses with the exclusion of non-English and/or unpublished studies (Hartling et al., 2017; Morrison et al., 2012). In this regard, some of the theses and dissertations (the Turkish ones) is retrieved from Turkish Council of Higher Education Thesis Center which is a formal data base for archiving the theses and dissertations prepared for the completion of Master's and Ph.D. programs and approved by an academic jury of universities in Turkey. (3) The studies were accepted for inclusion on the condition that they are empirical studies on investigating the effect of digital or non-digital games on learning achievement, and (4) those reporting such statistical values to calculate a standardized effect size as sample size, standard deviation, mean. In the case of lack of necessary data of any study for analysis, contact was made with the authors for missing information. The detailed information about the studies obtained during data collection process is given in [Figure 1](#) in accordance with the PRISMA Flow Diagram (Moher, Liberati, Tetzlaff, Altman, & Prisma, 2009).

We made sure that the studies included within meta-analysis had trial and control groups in order to obtain the standard effect size. The experimental groups represented the learning environments which used educational gaming method, and the control groups stood for the samples for which the gaming method was not applied. As shown in [Figure 1](#), we reached a total number of 1920 studies as a result of scanning the databases afore-mentioned. Out of these studies, 371 of them were excluded as they were duplications, and the abstracts of 780 of them were read and found to have irrelevant topics, thus were eliminated as well. 558 of the remaining 769 studies

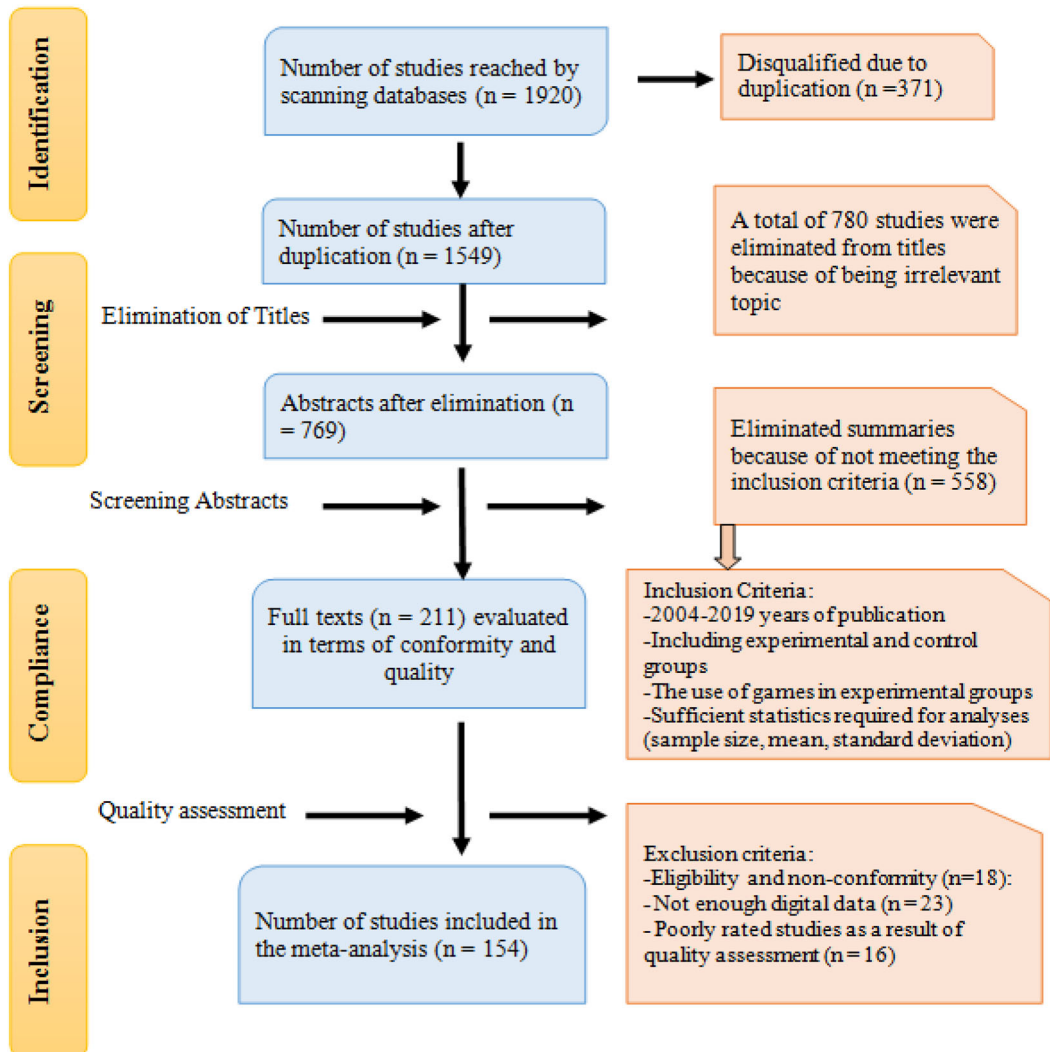


Figure 1. Flow chart for selection of studies.

were further disqualified from analysis as they did not meet the inclusion criteria. On examining the other 211 studies in terms of their suitability and quality, 18 studies were also excluded from analysis. Finally, within the context exclusion criteria, it was decided to exclude 23 of these studies as they did not include required data (x, ss, n) that are necessary to conduct meta-analysis. Because of the fact that such factors as bias, poor design, and missing/unsatisfactory findings negatively affect the quality of research studies and they are scientifically regarded as poor-quality, a further 16 of them were also excluded from the study. Hence we decided to include 180 effect size comparisons out of a total number of 154 publications compatible with the research problem and the inclusion criteria into meta-analysis. Some of these 154 studies were seen to have dealt with the academic achievement variable in more than one subject at the same time; and within some other studies the effect of educational games on academic achievement were understood to have been researched with different sample groups. Therefore, in accordance with the structure of meta-analytical research, the findings of each of those studies in question were separately taken into meta-analysis, and were included in the research as separate studies, so the number of studies was accordingly figured. In addition, the total sample size of the studies was identified to

consist 12800 participants from trial ($n=6493$) and control ($n=6307$) groups. All the studies with their effect sizes and moderator codings included in the meta-analysis are given in Appendix A.

Coding method

The studies included into the research were exposed to a coding process in order to make a comparison among the studies obtained from the literature search and to understand whether the studies are compatible with the inclusion criteria of meta-analysis. To this end, a coding form was prepared in order to ensure the reliability and validity of this research study (Bangert-Drowns & Rudner, 1991). This coding form was composed of two sections. In the first section, such information as study-codes, names of authors, publication years, courses in which games were used, students' levels, duration of implementations, kinds of games and types of achievement tests were used as data. In the second section, there were descriptive statistical values of the data required for calculating the effect size of the studies [(sample sizes (n), means ($\bar{X}$), standard deviation (sd)). For the reliability of the research, coding processes were separately examined by at least two independent coders. Upon any inconsistency, coding processes were reinitiated and analysis went on until an exact inter-rater agreement was reached. The Kappa statistics suggested by Cohen (1960) for determining the inter-rater reliability was preferred. This technique, developed in order to identify the degree of consistency (k) between two raters, takes values from -1 to $+1$ (Fleiss, 1971). These values are as: $k < 0,00$: poor, $0,00-0,20$: slight, $0,21-0,40$: fair, $0,41-0,60$: moderate, $0,61-0,80$ substantial, $0,81-1,00$: almost perfect agreement (Landis & Koch, 1977). To this end, it was ensured that the two different raters examined the data set and studied independently. The similarity between the codes of the raters is attached a great importance. As a result, the inter-rater reliability was found to be 0.92 according to Cohen's agreement statistics. Furthermore, with the calculation made according to the consensus percentage (consensus/(consensus + dissensus) $\times 100$) (Miles & Huberman, 1994) a 93% consistency was found. The two calculation values show that the coding process was reliable.

Data analysis

After identification of the data to be used for meta-analysis, the obtained data were coded and tabulated with Microsoft Office Excel program, the descriptive analyses were filtered and the relevant findings were calculated as percentage and frequency. In this research, the effect sizes of each study, heterogeneity test, and publication bias calculations were realized with Comprehensive Meta-Analysis (CMA) Version 2.0 (Borenstein & Rothstein, 1999) and MetaWin 2.0 statistical version of programs (Rosenberg, Adams, & Gurevitch, 2000). There are different coefficient categorizations made by various researchers for effect sizes which are calculated as meta-analysis currency. The interpretation of the effect sizes of the studies analyzed in this research was made according to Cohen (1992)'s levels ($0,20 - < 0,50$ small; $0,50 - < 0,80$ medium; $\geq 0,80$ large) and considering 0.05 significance level.

Heterogeneity and effect sizes

The heterogeneity test (Q test) can be explained as a statistical test showing the Chi-square distribution with $k-1$ freedom value. It is stated by many researchers to be statistically powerful when used in large samples (Gavaghan, Moore, & McQuay, 2000). The rejection of the Q test, also known as heterogeneity test, can be interpreted as the existence of more differentiation in the data than reflected to sampling errors. Another value developed as a complementary to the Q statistics is I^2 . This value gives a clearer result for heterogeneity (Petticrew & Roberts, 2006) and

shows the total variance rate of the effect size. However, the I^2 value is affected by the number of studies of the Q statistics. In the analyses, 25% shows low heterogeneity, while 50% and 75% show average and high heterogeneities respectively (Cooper, Hedges, & Valentine, 2009). Accordingly, the I^2 value calculated in this study is 86.152, thus it means high degree of heterogeneity. In absence of heterogeneity, the effect sizes are reported related to the literature and the meta-analysis is ended. While effect sizes show a heterogeneous distribution, the group differences are tested according to various variables (Hedges & Olkin, 1985; Rosenthal, 1984).

Effect size is accepted as the most important unit of a meta-analysis, and is known as a value reflecting the magnitude of difference between groups or correlation between variables (Borenstein, Hedges, Higgins, & Rothstein, 2009). Such formula as Cohen's d, Hedge's g, Glass's Δ are encountered in the literature for calculating effect size values (Lipsey & Wilson, 2001; McMillan & Foley, 2011). For the calculation of effect sizes, Hedges' g effect size index, which gives the pooled and standardized means difference between groups, was used in the present study (Hedges, 1981). For the analysis of the data, treatment effect kind of meta-analysis technique was utilized, which requires being so selective in terms of inclusion criteria. In fact, we cannot expect these studies included in the research to show the same widespread impact; the fact that the studies were conducted under different conditions could indispensably cause differences among the results. Thus, in identification of effect sizes during the present meta-analytical processes, analyses were conducted according random effects model (REM). In the REM, the main aim is the estimation of the mean of the effect sizes distribution. Thus, whether it has a small or large sample size, each study should be represented within the summarized effect, and the relative weights in the REM shows a more balanced distribution (Borenstein et al., 2009). Hedges and Vevea (1998) and Hunter and Schmidt (2000) suggest that the most important point when deciding the model in a study is the nature of the inference that is wanted to be made. The REM model is recommended for the inferences related to the population which is selected as sample of the studies included in meta-analysis (Field, 2003).

The reliability of the study

Several analyses were conducted for the reliability of the study. Some methods are suggested in order to ensure reliability in meta-analytical studies. In that, an attention is drawn in the literature on publishing studies reporting positive results more, and it is emphasized that published articles report larger effect sizes than non-published ones (Cheung & Slavin, 2016); thus preventing publication bias and language-bias is stressed as they might cause erroneous outcomes (Bulpitt, 1988). The fact that the studies included in meta-analysis are selected only from those published ones showing significance brings about the case of publication bias. Therefore, a fail-safe number test which is done to decrease or remove publication bias is calculated (Rosenthal, 1979). After the analyses made in the present research, it was calculated that if further 67083 studies on the effect of educational games on academic achievement were to be included in the analysis, the significance in question would decrease to zero. Given the number of the studies included in the analysis, it is clear that the obtained fail-safe number is a very high value. In this case, as they are too great number of studies to be reached, it can be suggested that publication bias does not affect the effect size in our meta-analysis (Cheung & Slavin, 2013). This finding shows that the meta-analytical processes in the present study are highly reliable.

Another way of ensuring reliability in meta-analytical studies is calculation of the normal-quantile plot graphed in Figure 2. For this, the studies included in the analysis should be within the two lines in the graph. In Figure 2, each study is shown with a dot, and it is seen that these dots are within two lines. This shows that the effect size levels of the studies are within a normal distribution and the distribution is within reliable (Rosenberg et al., 2000). Additionally, the fact that the studies included in meta-analysis are within the two lines as in Figure 2 can be

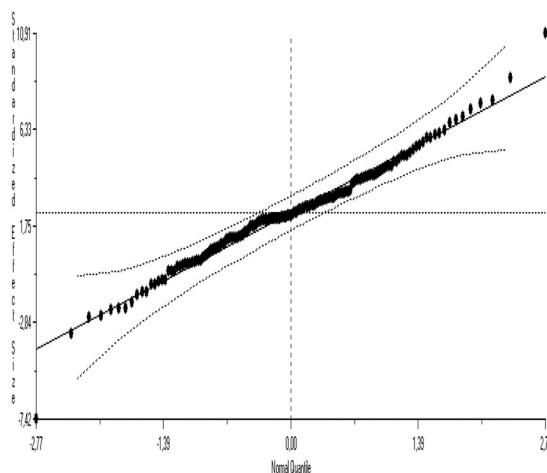


Figure 2. Normal quantile plot.

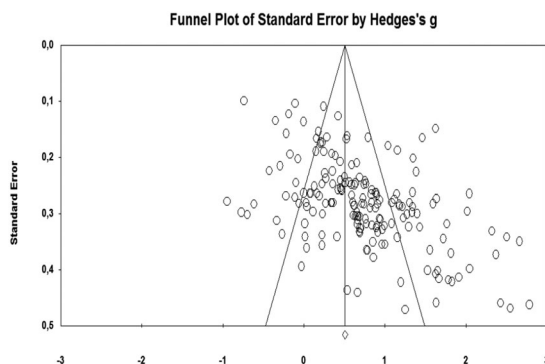


Figure 3. Funnel plot.

interpreted as that the obtained values for effect sizes do not show outliers, thus the data are suitable for calculating effect size. In this sense, a normal distribution means that the procedures to be used in meta-analysis will also be statistically suitable (Rosenberg et al., 2000). Thus, the analyses made for the use of educational games can be said to be highly reliable. Another most commonly recommended method for detecting publication bias in meta-analyses is funnel-plots (Duval & Tweedie, 2000). In a funnel plot, we can see effect size value in the X axis, while there are sample size, variance or standard error in the Y axis. The idea in the above graph is that as the sample size increases, the certainty in the effect size estimate will increase as well (Sterne & Harbord, 2004). In absence of bias, the graph is expected to be like an overturned symmetrical funnel as in Figure 3 (Sterne et al., 2011). In other words, when there is no bias, studies scatter to both sides of the vertical line, but when there is bias, studies scatter to one side and generally toward the bottom of the graph (Borenstein et al., 2009). In Figure 3, most of the 180 effect sizes seem to locate symmetrically upwards the graph, and scatter around both sides of the line. In this case, we cannot talk about existence of bias for the present study.

Findings of meta-analysis

Findings regarding the effect sizes of the studies on academic achievement

As can be seen in Table 1, as a result of the analyses made according to random effects model (REM); it was determined that with an overall effect size being “0.695” the academic achievement

Table 1. Distribution of homogeneous values, average effect sizes, and confidence intervals for studies included in the meta-analysis according to the effect models.

Model Type	n	Z	p	Q	g	I ²	% 95 confidence interval	
							Lower limit	Upper limit
FEM	180	27.661	0.000	1292.631	0.506	86.152	0.470	0.542
REM	180	13.762	0.000	187.441	0.695		0.596	0.794

df:179

is better in instruction with games than in instruction with traditional methods. When this effect size is examined, it is understood it is a significant positive value at a medium level according to the categorization of Cohen (1992). z- test calculation was found to be 11.688. Accordingly, with the obtained $p < 0.05$ result, it can be said that the analysis has statistical significance. As a result of the test of homogeneity, the Q statistical value was calculated as 185.936. With the Q statistical value 190.516, it is understood that it does not exceed the critical value of the chi-square distribution ($\chi^2_{(0.95)} = 190.516$). The overall effect sizes of studies included in meta-analysis are given according to the effect models in Table 1. The effect of use of educational games on academic achievement took a 0.695 value in REM, which means that the results are in favor of application of games in learning environments. In addition, we conducted analyses by splitting all the studies into different groups as well.

Moderator analyses

The overall effect sizes of the studies included in meta-analysis in accordance with the students' levels (of schooling) were identified. To this end, the studies to be analyzed were distributed into five groups as Preschool, Primary School, Middle School, High School and University. According to the analysis results in Table 2 below, the highest overall effect size is seen in the Preschool group with a 1.03 value, while the lowest overall effect size is in the High School group with a 0.43 value.

In Table 2, on examination of the intergroup heterogeneity test obtained from the results of meta-analysis, the $Q_B = 6.69$ value was detected. Accordingly the χ^2 distribution value was recorded to be as $\chi^2_{(0.95)} = 9.488$. Thus, when we grouped the studies included in meta-analysis according to the *students' levels* and examined their effect sizes; we found that there is no significant difference in terms of grade levels. This finding indicates that the academic achievement level in classes conducted with games does not show any change according to different educational stages.

In order to identify whether the overall effect size of the educational game applications change according to *duration of implementation*, the application durations within the studies were grouped as 1-4; 5-6; 7-8; 9-18 weeks and "Unspecified" and "Other." When we examine the analysis results of these six groups in Table 2, according to Cohen (1992)'s categorization, the lowest value is seen in the "Unspecified" group ($g = 0.52$), while the highest value in the "5-6 weeks" group ($g = 0.88$). These highest and lowest effect sizes are in the positive direction. On examining the intergroup homogeneity test, the value was found as $Q_B = 6.05$. The .05 significance level, the χ^2 distribution was identified as 11.070. Accordingly, it was determined that when we grouped the studies according to their durations of application and looked for the intergroup effect size ($Q_B = 6.05$; $p > 0.05$), there was no significant difference. Thus, it can be suggested that the academic achievement in the classes conducted with educational games does not change according to duration of implementation.

In order to determine whether the overall effect size of game applications change according to the *school subjects* of the studies included in meta-analysis, the subject areas were divided into 5 groups as; Science, Social Science, Math, Computer and Foreign Language Courses. According to

Table 2. The overall effect sizes of the studies according to moderator analyses.

Md	Groups	Effect size and 95% confidence interval				Test of null		Heterogeneity		
		n	g	Lower	Upper	Z-value	P-value	Q-value	df	P-value
Students' Levels	Preschool	8	1.03	0.65	1.42	5.29	0.00	6.69	4	0.15
	Primary	47	0.62	0.45	0.80	6.96	0.00			
	Middle	95	0.73	0.58	0.87	9.97	0.00			
	High School	13	0.43	0.12	0.75	2.68	0.01			
	University	17	0.76	0.45	1.07	4.76	0.00			
	Tot. Betw. Overall	180	0.69	0.60	0.79	14.12	0.00			
Durations of Implementation	1–4	60	0.64	0.47	0.81	7.38	0.00	6.05	5	0.30
	5–6	29	0.88	0.64	1.12	7.17	0.00			
	7–8	25	0.80	0.52	1.07	5.71	0.00			
	9–18	32	0.61	0.37	0.85	5.07	0.00			
	Unspecified	22	0.52	0.25	0.80	3.68	0.00			
	Other	12	0.84	0.56	1.12	5.87	0.00			
	Tot. Betw. Overall	180	0.70	0.61	0.80	14.36	0.00			
School Subjects	Science	43	0.85	0.66	1.05	8.43	0.00	26.91	4	0.00
	Social Sciences	29	0.53	0.34	0.73	5.33	0.00			
	Math	65	0.63	0.46	0.80	7.13	0.00			
	Computer	5	−0.12	−0.48	0.25	−0.63	0.53			
	Foreign Langs.	38	0.87	0.70	1.07	8.47	0.00			
	Tot. Betw. Overall	180	0.66	0.57	0.75	13.98	0.00			
Class Sizes	Small	70	0.87	0.71	1.04	10.29	0.00	13.17	2	0.00
	Medium	82	0.68	0.55	0.82	9.79	0.00			
	Large	28	0.37	0.15	0.58	3.28	0.00			
	Tot. Betw. Overall	180	0.68	0.59	0.78	14.12	0.00			
Kinds of Games	Digital	83	0.47	0.33	0.60	6.76	0.00	20.07	1	0.00
	Non-digital	97	0.90	0.77	1.03	13.41	0.00			
	Tot. Betw. Overall	180	0.69	0.60	0.78	14.34	0.00			
Types of Tests	RT	139	0.71	0.60	0.82	12.43	0.00	0.32	2	0.85
	ST	37	0.64	0.41	0.86	5.57	0.00			
	U	4	0.71	−0.14	1.57	1.64	0.00			
	Tot. Betw. Overall	80	0.70	0.60	0.79	13.71	0.00			

Table 2, the highest effect size is seen in Foreign Language courses with a 0.87 value, while the lowest effect size is observed in Computer courses with a −0.12 value. The intergroup homogeneity test value was detected as $Q_B=26.91$. Since the Q_B statistical value ($Q_B=26.91$, $p < 0.05$) is bigger than the critical value of the χ^2 distribution ($\chi^2_{(0.95)}=9.488$), it has a heterogeneous distribution. Thus, when we looked at the intergroup effect size, we found a significant difference among the groups. It is clear that the academic achievement in the classes conducted with games changes according to different subject areas (courses).

In order to determine the impact of the *sample sizes* of the studies included into analysis on the overall effect size, the studies were grouped as small sample (less than 50), medium sample (50–99) and large sample (100 and over) studies. The analysis results of the sample groups show that the highest effect size is of the small class sizes, while the lowest effect size is of the large sample studies. Additionally, the studies were further divided into 2 groups as those conducted with digital games and those with non-digital games in order to determine the impact of the *kinds of games* used in the studies on the overall effect size. And it is obvious that the effect size of the non-digital kinds of games is higher.

Upon the examination of the intergroup homogeneity test in terms of the *class sizes* of the studies, the $Q_B=13.17$ value was identified while the critical value of the χ^2 distribution was ($\chi^2_{(0.95)}=5.991$). We saw that there is a significant difference in terms of class sizes. Similarly, when we examined the intergroup homogeneity test in terms of the *kinds of games* used, we saw that the Q_B value (20.07) is bigger than the critical value ($\chi^2_{(0.95)}=3.841$). Therefore, the academic achievement in the classes conducted with games shows changes according to both class size and

kinds of games used. Moreover, we looked for whether the effect size is moderated by the *type of academic achievement tests* (Researcher-made Tests, Standardized Tests, Unspecified), and found that the Q_B value (0.32) is smaller than the critical value of the χ^2 distribution was ($\chi^2_{(0,95)}=5.991$). Thus, no significant difference was identified in the effect sizes in terms of the types of academic achievement tests used; which means the academic achievement in the classes conducted with games does not change according to different types of achievement tests.

Methodology of the qualitative part (meta-thematic analysis)

Qualitative research design

A qualitative dimension (meta-thematic analysis) was also included in the study in order to further probe the subject-matter in more details, to supply a more comprehensive information and to provide the study with rich data (Batdı, 2017a). In the meta-thematic analysis method, it is aimed to bring together and compile from some selected qualitative studies the participants' views regarding the research subject (Batdı, 2017b). When considered from this point of view, what we mean to say about the meta-thematic analysis is that it is a method involving the process of reviewing and reassessing the themes formed from participants' views of various thematic analyses. Accordingly, within the meta-thematic analysis process, the interpretation of the data obtained via document analysis was made. Document analysis is a method frequently used to extract data, analyze them and draw conclusions by accessing documents and resources within the targeted context (Çepni, 2010). The findings obtained from these analyses were presented quantitatively so as to be interpreted and expounded integrally.

Data collection and analysis

In the qualitative dimension, which enriches the data set and expands the scope of the research, 7 studies out of those included in the meta-analysis and containing qualitative data were selected. Thus, the data-base and publication dates of these studies are the same; however, the fact that those studies selected for meta-thematic analysis are of qualitative dimension and based on participants' perceptions were the qualities looked for in the first place. The searched qualitative studies on educational games ($n=7$) were analyzed with MAXQDA-11 program. The data were analyzed in accordance with content analysis. The main purpose of content analysis is to arrive at some concepts and relationships which could expound the data obtained within the scope of a research-study. To this end, conceptualizations are made from the data of the studies, similar ones of these concepts are aggregated and coded under certain themes, and then after necessary arrangements, the results are interpreted (Yıldırım & Şimşek, 2013). In parallel with this, in this research, the themes and codes of similar nature from the studies on educational games were rearranged within a certain accord and order, and thus qualitative findings were obtained, then these findings were presented as models. Moreover, while presenting the statements in the studies from which the themes and codes were taken, they were shown within text with the intent of corroboration, justification and most importantly for the purpose of ensuring reliability. During the presentation of these quotations, no modification was done over the statements, they were taken as they were. The presentations of the direct quotations were indicated with study-codes and page numbers (i.e., in this numeral expression "422970-G-92"; 422970 code refers to the number of the study, "G" code refers to traditional, non-digital games and "92" refers to the page number of the study. And, in the numeral expression "8-T-1103"; "8" stands for the study-number, "T" code represents technology-based (digital) games and "1103" shows the page number of the quotation).

Reliability of the meta-thematic process

As a necessity for the procedures of ensuring the validity and reliability of the qualitative dimension of this study, it was made sure that the inter-consistency and meaningfulness of the codes and themes constituted an integrity for providing coherence of the findings. It is expected that well-prepared studies have already calculated the agreement among raters (McHugh, 2012). What's more, Cohen's Kappa statistic, which gives the coders' agreement, was used in order to ensure the inter-rater reliability (Yıldırım & Şimşek, 2013). As stated before, the intervals of agreement values are .61-.80 "substantial agreement"; .81-1.00 "almost perfect agreement" (Viera & Garrett, 2005). In the present study, the agreement value intervals were found to be between .722 and .802 as "almost perfect agreement" (Appendix B).

Meta-thematic results regarding the efficiency of educational games

In this part of the study, a meta-thematic analysis was adopted as a complementary support for the meta-analytical process. As mentioned earlier, a meta-thematic analysis, with a document analysis nature, is based on a detailed analysis of all participants' views about the same certain topic of various qualitative studies, and modeling them by aggregating into themes and codes. To this end, to consolidate the academic achievement theme dwelled on in the meta-analysis, the theme "Cognitive Contributions of Educational Games" (Figure 4) was formed. In addition, we decided to thematize two more titles with a more general sense as "Drawbacks of Educational Games" (Figure 5) and "Suggestions" (Figure 6), thus forming 3 different themes in total.

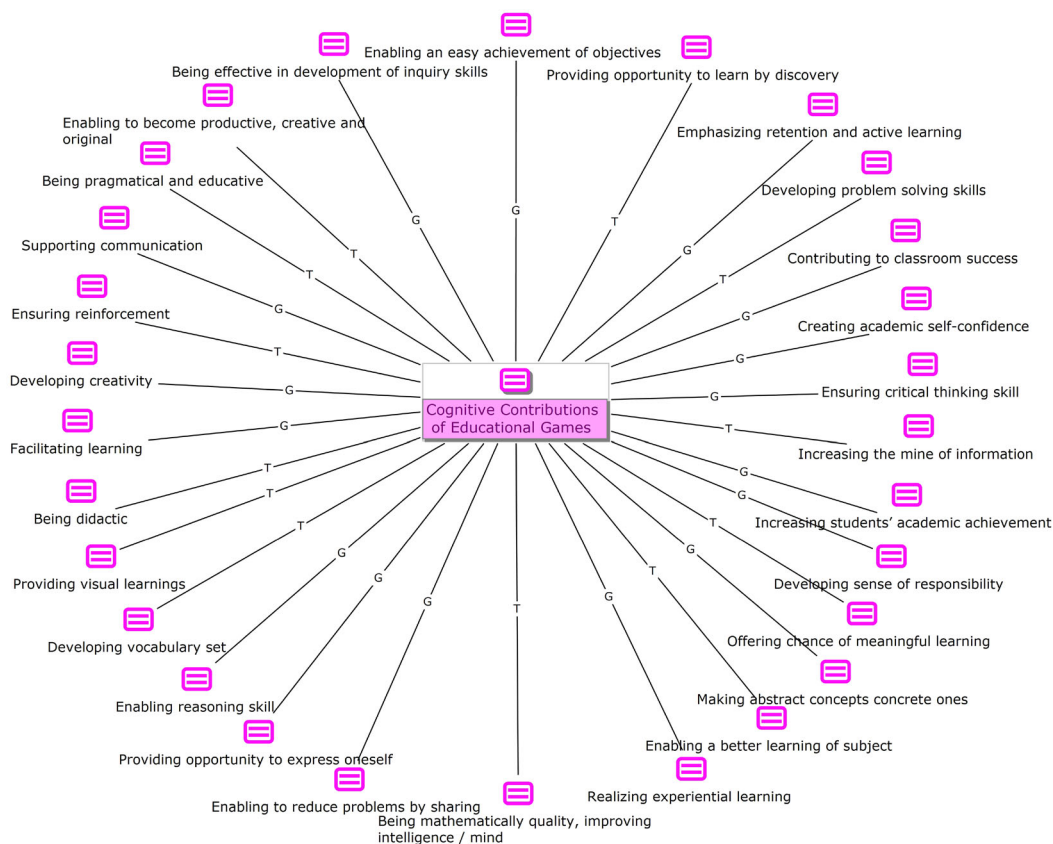


Figure 4. Cognitive contributions of educational games.

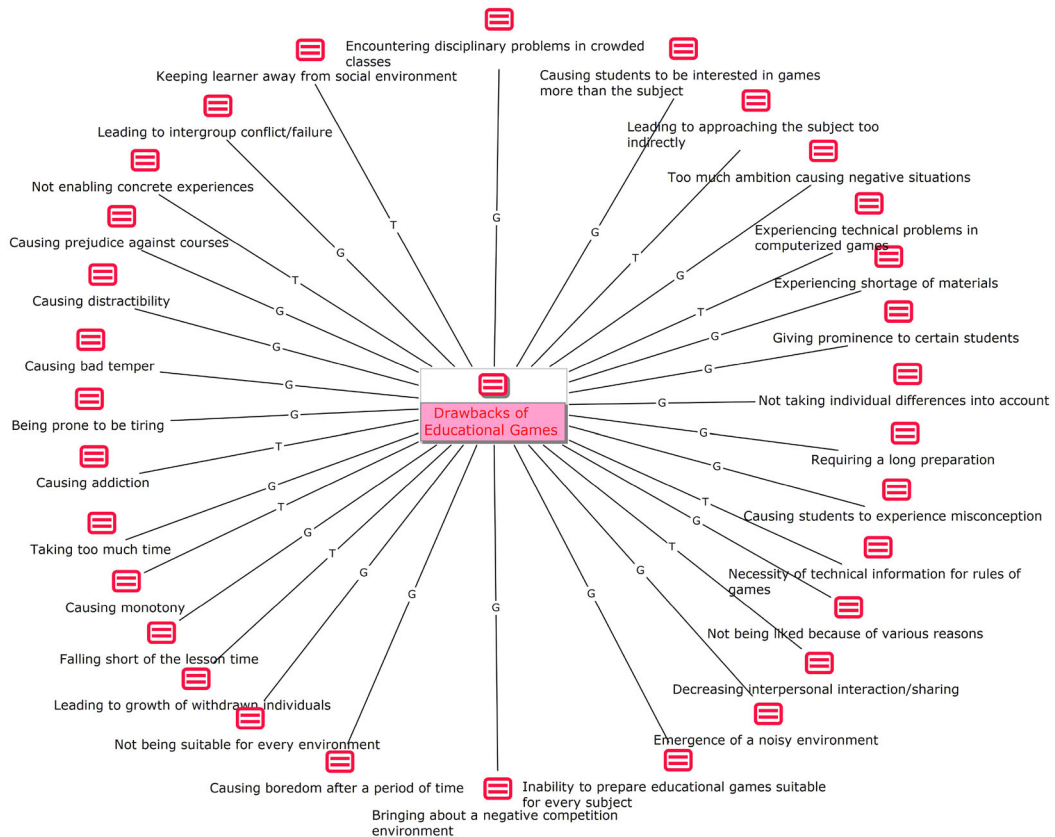


Figure 5. Drawbacks of educational games.

Meta-thematic findings regarding cognitive contributions of educational games

Thus, the contributions the applications of games make to the participants' cognitive domains are modeled in Figure 4 with the relevant theme title.

In Figure 4, some of the codes regarding the use of games in teaching-learning process can be expressed as "Providing opportunity to learn by discovery, Developing Problem Solving Skills, Enabling to become productive, creative and original, Contributing to classroom success, Being mathematically quality, improving intelligence/mind, Providing opportunity to express oneself, Developing sense of responsibility, Emphasizing retention and active learning, Being effective in development of inquiry skills." Within the context of these codes, the following statements were taken into consideration while forming the relevant codes: from the study coded 8-T-1103 "It affects problem solving skills. The process of trying to find various solutions for the problems they encounter, or for what is wanted and unwanted in the game affects this process of problem solving skill development directly"; from the study coded 330171-T-98 "I think it is more practical. And we happen to develop our brains by solving problems without using papers. Most of games are fun and mathematical." Moreover from the study emphasizing creativity and coded as 422970-G-92 "... our creativity develops then and we dream of being in the classroom, we think of experiencing some problems and try to find practical solutions to these problems"; from another study mentioning critical thinking "Games enable us to look critically" (422970-G-92); and a study emphasizing inquiry skills and retention "...when taking retention of knowledge into account, it is the method I prefer" (456123-G-36), "It has positive effects, I myself made an educational game application and as a result it contributed to the students' inquiry skills. It made

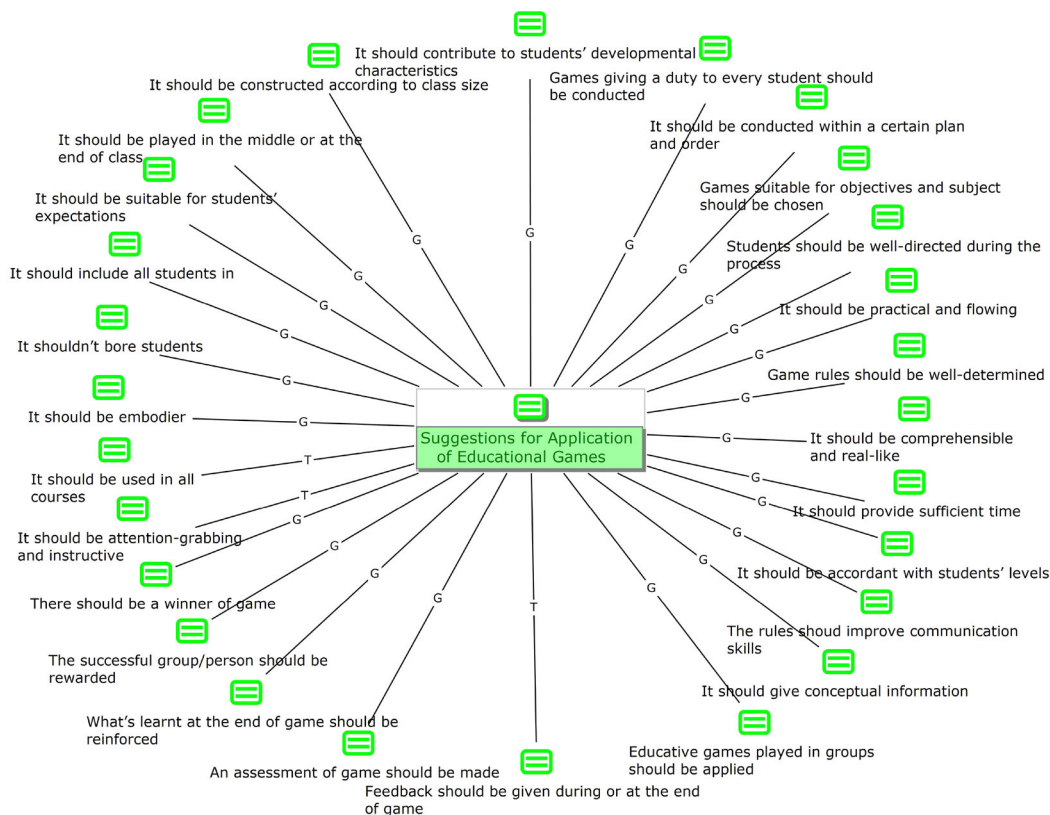


Figure 6. Suggestions for application of educational games.

the subject I taught more permanent” (476575-G-89). In addition to these all, when we examine the model, we can say that the relevant applications may contribute to the participants’ cognitive domains in many ways such as sharing of problems, increasing their vocabulary set, forming of visual learnings, etc.

Meta-thematic findings regarding the drawbacks of educational games

As a result of the meta-thematic analysis of the relevant studies, it was observed that there are drawbacks that come forth with use of games in learning environments. These negative aspects of educational games were synthesized with a model in Figure 5 with a related theme title:

In Figure 5, some of the codes regarding the drawbacks of using games in teaching-learning process can be expressed as “Decreasing interpersonal interaction/sharing, Bringing about a negative competition environment, Causing students to be interested in games more than the subject, Emergence of a noisy environment, Experiencing technical problems in computerized games, Causing monotony, Not enabling concrete experiences.” In the formation of the relevant codes, some statements taken as references from the studies in which digital games were conducted are especially attention grabbing. From the relevant studies particularly the one dwelling on the drawbacks of digital games and emphasizing a comparison indicates some important points (coded as 8-T-1103) “Computer games may lead to addiction in children ... Educational computer games affect interpersonal interaction in a negative way, and make children withdrawn, bored, diffident individuals. ... Children spend so much time with computers that they grow being disconnected from the real world, thus instead of giving education with computers by taking them away from

social environment, turning to traditional games would rather include children into the process.” Moreover, the statements quoted from the study coded as 330171-T-101 can be references for the drawbacks of digital games: “The computer I was working on was once frozen. In this case, I had to go on with another one” and from the study coded as 8-T-1103 “In the traditional game approaches, child takes place in the game personally, he/she shapes and designs the game himself/herself, but rules are preset beforehand in computer games thus they cannot gain concrete experiences.” At the same time, the study coded as 456123-G-36 reflects the case clearly while dwelling upon the drawbacks of non-digital games “I think there are activities better than games. There is a lot of waste of time. It requires a long preparation. Students get interested in games more than the subject itself.”

Meta-thematic findings regarding suggestions for application of educational games

The studies that were included in meta-analysis and meta-thematic analysis reported participants’ views to be taken into consideration while implementing game activities for instructional aims. These views were reevaluated and brought together as suggestions for application of games with a model shown in Figure 6:

In Figure 6, it can be seen that the various suggestions from different studies for application of educational games are modeled with codes. Some of these codes from the prominent studies including both digital and non-digital games are as follow: “It (game) should be attention-grabbing and instructive, Games suitable for objectives and subject should be chosen, It should be accordant with students’ levels, What’s learnt at the end of game should be reinforced, It should be used in all courses.” These statements quoted from the 476575+-G-92 coded study emphasizing the suggestions for the use of non-digital games in learning environment are worth paying attention “It should be an educational game in which all students can get involved and which can draw students’ attention. Otherwise, the game cannot attain its aim and the class may fall apart” and “Games should be arranged within the scope of targeted objectives. They should be compatible with students’ growth levels, and simple and comprehensible. The materials to be used in the game should be durable so that they could be played with again and again.” One study coded as 435990-T-81 examining the effect of digital game applications highlights the fact that games can be used in all courses “Educational games should be utilized in every course. And they help students in learning, the students can consolidate what they learn through games.” Moreover, it is further suggested that when both non-digital games and digital games are used in learning environment, such significant points as concretization, use of time, inclusion of all students into the process, specification of game rules well in advance, etc. are said to be paid attention.

Discussion

Academic achievement

This research study embodies meta-analytic and meta-thematic evaluations of use of games during the instructional process. Within this framework, in accordance with the purpose of the study, a meta-analytic examination was first resorted to exactly establish the effect of game activities on academic achievement. The result of meta-analysis showed that educational games have a positive effect on academic achievement and this effect is at medium level ($g = 0.695$). This result lays bare the efficiency of educational games in increasing learners’ academic achievement. Likewise, the relevant studies which revealed the result that educational use of games affects learning in a positive way and is effective in boosting achievement also support the present study (Chen, Tseng, et al., 2018; Chian-Wen, 2014; Liao et al., 2010; Tokac et al., 2019). The classes conducted with educational games are thought to contribute to enhancing student success since they encourage students to improve their critical thinking, reasoning and problem solving skills,

they concretize abstract concepts; and as they ameliorate students' physical and mental abilities and they also contain entertaining elements (Gürpınar, 2017). In the literature, it is stated that use of games for instructional purposes makes students enjoy learning and feeds their self-confidence, imagination and power of creative thinking; and enables class integrity by decreasing jealousy and anxiety, and directs to right information by providing instant feedback (Alici, 2016; Işık, 2016). These can be regarded as factors strengthening student achievement. Moreover, being a learner-centered technique, instructional games make students have amusing time, render courses more pleasant and productive thus directing all attention to the subject matter and triggering motivation. This brings along effective learning together with success.

While there are studies concluding that educational games increase academic achievement, there are also some studies reporting no significant increase in achievement at all. The reason for various studies' presenting different results in this respect may be that games are applied in different ways as there are various types of games for educational purposes (the ones requiring no digital technology and digital games such as computer games or video games designed specifically for instruction), and that they may not have content suitable for students' grade level; and the degree of impact of the application time on students could be a reason as well. Another reason for the reported differences might be teachers' organization and administration of the process in totally different ways. In addition, students' motivation and attitudes toward the course may have influence on obtaining different results from the studies. Accordingly, it can be suggested that the content should be properly and fairly designed in harmony with the target sample and the courses should be well-planned in order to accomplish.

On the other side, moderator analyses were conducted within the meta-analytic evaluation, and we tried to identify whether the efficiency of games in the teaching-learning process changed in terms of grade levels, durations of application, subject areas, class sizes and kinds of games. In order to ensure the reliability of meta-analysis, a calculation of publication bias was carried out, and when we looked whether the effect sizes of the studies showed a normal distribution, the results indicated that the analyses made were reliable. As a result, it was determined that the studies in which use of games for instruction is researched changed in terms of the subject areas which used games, their class sizes and kinds of games used for instruction.

School subjects

In terms of school subjects (courses), the highest effect size was observed in foreign language courses ($g=0.87$), preceded by Science ($g=0.85$) and Mathematics ($g=0.63$) respectively; while the lowest effect size was found to be in computer courses ($g=-0.12$). This result is in line with Mayer (2019)'s review on computer games in education, which suggests that science, mathematics and second language learning are three promising areas in which games could be more effective than traditional media. In a similar fashion, Randel et al. (1992)'s review on effectiveness of educational games puts forward Mathematics and Language Arts as two distinct subject areas in which use of games for instruction proves better than conventional teaching. As to the fact that use of games for instruction shows its highest effect on achievement in foreign language courses, it would not be an exaggeration to claim that gaming pays off best in language learning endeavors (Gros, 2007), as game play provides the social context and interaction purported by the social constructivist theory and modern communicative theories for development of language skills. The fun element found in games is thought to be the incentive which triggers simulated communication among language learners (Doğan, Batdı, Yıldırım, & Dönder, 2012). There is accumulating evidence to support the usefulness of both digital and non-digital gaming in foreign language learning. It is reported that participation in game-based language learning activities ensures literacy attainment (Mifsud, Vella, & Camilleri, 2013), contributes to vocabulary acquisition (Ranalli, 2008), fosters communicative competence and performance (Peterson, 2009; Zhu, 2012), plays

role in long-term language maintenance and learner autonomy (Godwin-Jones, 2014), serves for experiential learners, offers enriched and meaningful language input and enables to use language so as to obtain nonlinguistic outcomes (Tomlinson & Masuhara, 2009). However, gaming in science education has also been reported to yield promising results as games provide simulated scientific contexts where students could experience relevant phenomena which they would not be able to do in classroom environments (Cheng, Chen, Chu, & Chen, 2015; Magnussen, 2014). In their review of 50 studies on the use of educational computer games in teaching of Science-related school subjects during the period of 1999-2009, Svingby and Nilsson (2011) state that in terms of assessing learning impacts, computer games make a positive contribution to Science learning; and they underline the conclusion that if designated in line with the criteria which are suggested in educational theories of game play, games furnish learners with the opportunities of applying scientific inquiry skills. Moreover, the utilization of games, particularly video games and serious games, is shown to contribute to the comprehension of social facts in education of social sciences (Cuenca-Lopez & Martin-Caceres, 2010; Jancic & Hus, 2018). It is stressed that school subjects in which content is highly specified and objectives are exactly determined could presumably be more efficient for gaming interventions (Randel et al., 1992).

Class size

In terms of class sizes, it was understood that the overall effect size of the studies with small class sizes (less than 50) was at a large level ($g=0.87$). It is clear that gaming in classes with small number of students turn out to be more efficient as most of the negativities reported to have been encountered during implementations could be minimized with a stronger conduct of classroom management. The research on educational interventions in terms of class size show unanimously that learning in classes that are not large yields more fruitful results, and no major disciplinary, physical or instructional problems are experienced (Ijaiya, 1999; Khan & Iqbal, 2012; Yaman, 2009). In terms of non-digital games, small class size provides a higher level of teacher and student participation within a casual game atmosphere whereas large classes necessitate more maintenance and direct teacher control (Sugar & Sugar, 2002).

Kinds of games

And last but not least, the highest overall effect size in terms of kinds of games conducted was seen to be in non-digital games ($g=0.90$). As for the non-digital games showing a higher effect size, we can draw the conclusion, with a cautious optimism, that non-digital games could be more efficient in learners' development and learning as they supply more opportunities for interaction among peers, a wider spectrum of activity choices, more flexibility of design and content, and a variety of platforms for playing. Some research findings in this regard exist in the literature reporting that non-digital games might provide more benefits than digital games (Edwards, 2014; Ernest et al., 2014; Tomlinson & Masuhara, 2009). However, this does not mean that we underestimate a plethora of merits already obtained from the educational digital gaming research (Çiftci, 2018; Clark et al., 2016; Felicia, 2009; Fletcher & Tobias, 2006; Granic et al., 2014; McGonigal, 2011; Posso, 2016; Prensky, 2006; Ritzhaupt et al., 2014; Shaffer, 2008; Shapiro, 2014; Tobias, Fletcher, & Chen, 2015; Yang, 2012). Furthermore, Ritzhaupt et al. (2014) warn in this sense about mistaking digital games for non-digital ones, and call for a caution about differentiating their effects on learning. And Granic et al. (2014) point to the fact that there has been a dramatic change in the nature of digital games over the last decade as they have become more complex and realistic in nature, thus offering more potential with regard to improvements in childhood and adolescence mental health and well-being.

Educational stages, implementation periods and assessment of achievement

The duration of exposure to educational game interventions is reported to be a determinant of what to be learnt and experienced (Clark et al., 2016; Svingby & Nilsson, 2011). Thus, we wondered whether the academic achievement in the classes conducted with educational games changes according to various implementation periods; however, we found no significant difference in terms of time periods allocated for teaching through games. Further findings of the moderator analyses indicate that the academic achievement level in classes conducted with games does not show any changes according to different educational stages and assessment types of academic achievement. The results our meta-analysis regarding the types of achievement tests (researcher-made, standardized and unspecified tests) are in line with those reported by Clark et al. (2016), who did not find any evidence of a relationship between assessment type (i.e., preexisting normed instrument, modification of a preexisting instrument, author-developed instrument) and effect size. Therefore, these authors suggest that as long as they report reliability and validity, researchers should be welcome to use instruments whether modified versions of standard ones or researcher-made ones. It is also clear that incorporation of educational games into the teaching-learning process produces promising learning outcomes in all stages of education.

Cognitive contributions of games

As a complementary evaluation to the meta-analysis results within the scope of the study, meta-thematic analyses were made by forming such codes and themes as the effects of educational games on different dimensions. Accordingly, meta-thematic analysis enabled to reinterpret and restate the relevant themes and codes formed from the raw data of participants' comments (Batdı, 2019) on educational games. To further clarify the achievement theme dealt with in meta-analysis, a theme detailing the contributions of educational games to cognitive domain was formed. When we reevaluated the codes collected under this theme, it was revealed that educational games are effective in learner achievement, they improve creativity and mental functions, facilitate learning process and provide a meaningful learning (Amory, Naicker, Vincent, & Adams, 1999; Kaya & Elgün, 2015; Şahin, 2016). In addition to helping to develop such skills as imagination, synthesis, problem solving, inquisition, reasoning and critical thinking, games have been found to engage students' attention and enable retention of knowledge (Cicchino, 2015; Gürpınar, 2017; Hwang et al., 2012; Kinzie & Joseph, 2008). Using games for instructional purposes offers an opportunity for learners to learn experientially, and it is efficient in fostering students' sense of responsibility as well. It has also been reported that educative games concretize abstract concepts that are complex and hard to understand and turn them into entertaining ones (Aksoy, 2014; Peirce, Conlan, & Wade, 2008; Yazıcıoğlu, 2017; Yıldırım, 2016). In parallel with these all, the relevant literature has proved that games in educational settings make teaching more effective by creating meaningful and perpetual learnings, and they enhance students' attitudes, interest, motivation, regulation and engagement (Bai, Pan, Hirumi, & Kebritchi, 2012; Little, 2015). The fact reiterated by both the qualitative studies included in this research and those existing in the literature that games provide cognitive gains support the relevant finding of our meta-analysis, which reveals the significant positive effect of both digital and non-digital of games compared to traditional teaching.

Drawbacks

Together with their above-mentioned merits at cognitive level, educational games have been reported to have some drawbacks, too. Some of these negative aspects are purported as that such problems as disorder, noise, objections, criticisms and challenge can be encountered during the

application of educational games in crowded classes; students may get interested in the game itself more than the subject-matter, thus teachers might not ensure classroom management (Bayram, 2015; Gürpınar, 2017). As to especially computer-based games, they may cause addiction and bad temper, lead to some psychological problems like making individuals withdrawn and alienating them from social environment (Güntepe & Usta, 2017; Gülay, 2008; Van Eck, 2006). Besides, the fact that students do not want to get out from game mode and that games are not kept under control could cause games to deviate from their aim (Gürpınar, 2017; Özdemir, 2018). And more about digital games, it has been revealed that spending long time in front of computers brings about some health problems such as eye strain, headache, backache, sleep disorders (Topçu, Küçük, & Göktaş, 2014). It is also established that games may create negative competition environment, lessen interaction and sharing among individuals, and make students develop prejudices against courses (Güntepe & Usta, 2017; Yıldırım, 2016).

Suggestions for implementation

In order for educational games to be implemented in an effective and fruitful way, the problems discovered within the meta-thematic analysis are supposed to be resolved. Thus, a “suggestions” for implication theme was formed within the present study. When we reassess the relevant codes gathered under this theme, it can be suggested that games for educational use should be organized or designed suitably for class levels, learners’ expectations, and contribute to developmental characteristics of students. Additionally, during the application of games, time should be well-arranged and managed, students should be guided and learning environment should be prepared for game activity. Feedback should be provided during or at the end of games, thus reinforcing those newly learnt. It is important to make an overall evaluation of the game activity and to accordingly reward the successful learners. In order to sustain the application success, it is recommended to choose games fitting instructional objectives and the subject to be taught; furthermore a proper predetermination of game rules, conduction of games within a certain plan and order; designing learner-centered, attention-grabbing, understandable, down to earth, practical and flowing instructional games, whether digital or not, are all crucial points to be taken into account if their success-related objectives in question are desired to be accomplished.

Limitations

The effect-size findings of the study should be carefully evaluated together with its limitations. One significant limitation of the study is that no discrimination was made in terms of the research design types of the studies, as most of them used a quasi-experimental design while some were randomized controlled trials. Another limitation is that the studies were not grouped according to their publication types (as journal articles and unpublished dissertations). In this sense, it is suggested by Cheung and Slavin (2016) and Bakker et al. (2019) that effect sizes might be larger for published articles, quasi-experiments and researcher-made measures. What’s more, the games implemented within the studies were only grouped and coded as digital games and non-digital games, but were not specified with their sub-category. In fact, it is thought that a further categorization of digital games and non-digital games would provide a better reference for their effectiveness in teaching. And it should be noted that apart from academic achievement games could be effective on learners’ attitudes, motivation, behavior formation and affect.

Conclusion and implications

Regarded as a complementary part of human learning and development, playing games has long been on the agenda of educational research owing to the possible benefits it could provide for

teaching of various school subjects. Thus, we tried to make a research synthesis of the relevant studies conducted within the last fifteen years. As a result, it was found out within the meta-analysis of 154 research studies (with 180 effect sizes and an overall sample of 12800 participants) that educational use of both digital and non-digital games has a positive effect on academic achievement. In addition, a more detailed analysis of the overall effect was made with such moderating variables as educational stages, durations of implementation, subject areas, class sizes, kinds of games and types of achievement tests. One significant conclusion that can be drawn from this research synthesis is that it yields good results to take a game-based approach to teaching in one way or another. Given the research literature it would be appropriate to propose that even if it will not be a total transformation of our education into a game-based system like the *Quest to Learn* and the *Playmaker Schools* in the USA (Miller, 2012), we already have justified reasons to make games, whether digital or not, an indispensable and integrated part of our curricula.

There are certainly some implications for future research. The general propensity within the research on educational digital games is to investigate their effects on learning and achievement with a comparison to conventional instruction methods; however, we believe that some future empirical research which are designed to test the efficiency of any digital game with its non-digital equivalent serving the same purpose of instruction could probably present us a bordered framework for differentiating between the effectiveness of these two. Although playing games is a characteristic unique to all human beings, conduction of some cross-cultural studies might provide a specification of preferences for, qualities and resulting efficiency of educational games within various national contexts. Moreover, there is an increasing consensus among researchers of the field over designing educational games in which targeted learning objectives and their assessment are embedded in games. Some additional implications for future research are the variables not analyzed in detail in the present study but mentioned in limitations of the study; thus they could be suggested to the researchers for their moderating effects. And last but not least, we witness a negligence of non-digital games within the relevant research, thus emphasizing the need for conducting more empirical studies that could give an outline of their impacts on various components of learning in terms of diverse categories of non-digital games.

Note: The studies included in the meta-analysis are shown with one asterisks (), while the reference to those included in the meta-thematic analysis is made with two asterisks (**). The studies included both in the meta-analysis and meta-thematic analysis are referred with (*/*).*

Notes on contributors

Tarik Talan, PhD, is a fulltime assistant professor at Gaziantep Islam Science and Technology University, Turkey. His research areas include blended learning, distance learning, human-computer interaction and use of social media in education.

Yunus Doğan, PhD, has taught English at various K12 grades, and currently works at a state university in eastern Turkey. His research interests include technology-enhanced learning, individual differences in SLA and virtual learning environments.

Veli Batdı, PhD, is an associate professor of curriculum and instruction at a state university in Turkey. He specializes in using various qualitative and quantitative data analysis programs in his studies; and currently conducts meta-analysis, thematic analysis and meta-thematic analysis within education research.

ORCID

Yunus Doğan  <http://orcid.org/0000-0002-5038-3404>

Veli Batdı  <http://orcid.org/0000-0002-7402-3251>

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Appendix A. The meta-analysis data collection form

Author and year	N	Sample size	Grade/Class	Course/Field	Weeks	Kinds of games	Effect size
Abdelhafez (2016)	40	Small	High School	Math	6	Digital	0,409
Adanaş (2019)	36	Small	Secondary	Math	5	Non-digital	2,784
Akın and Atıcı (2015)	49	Small	Primary	Math	Unspcfd	Digital	−0,059
Akinsola and Animasahun (2007)	146	Large	Secondary	Math	3	Digital	0,524
Aksoy (2014)	40	Small	Secondary	Math	14	Digital	−0,336
Alici (2016)	60	Medium	Secondary	Science	6	Non-digital	1,118
Altınbulak et al. (2006)	54	Medium	Primary	Social Sciences	Unspcfd	Non-digital	1,763
Aremu and Adebagbo (2019)	50	Medium	Primary	Math	Unspcfd	Digital	0,819
Arslan (2016)–1	60	Medium	Secondary	Math	4	Non-digital	0,419
Arslan (2016)–2	60	Medium	Secondary	Math	4	Non-digital	0,832
Arslan (2016)–3	60	Medium	Secondary	Math	4	Non-digital	0,457
Ashraf, Motlagh, and Salami (2014)	24	Small	University	Foreign Langs.	15	Digital	−0,029
Aslan (2014)	49	Small	Primary	Math	4	Digital	−0,059
Aşçı (2019)	43	Small	Secondary	Foreign Langs.	6	Digital	0,228
Atay (2007)	29	Small	High School	Foreign Langs.	8	Non-digital	0,859
Atay (2018)	41	Small	Primary	Social Sciences	4	Non-digital	0,899
Aziz, Subiyanto, and Harlanu (2018)	72	Medium	Secondary	Foreign Langs.	14	Digital	0,755
Babaandaç (2013)	61	Medium	Secondary	Science	5	Non-digital	2,662
Bai et al. (2012)	437	Large	Secondary	Math	18	Digital	−0,739
Batdı (2013)	68	Medium	Primary	Foreign Langs.	15	Non-digital	0,561
Bayırtepe and Tüzün (2007)	51	Medium	Secondary	Computer	2	Digital	−0,615
Bayram (2015)	29	Small	Secondary	Social Sciences	Unspcfd	Non-digital	0,034
Benoit (2017)	60	Medium	Secondary	Foreign Langs.	6	Digital	0,469
Biriktir (2008)	41	Small	Primary	Math	8	Non-digital	0,775
Bozoğlu (2013)	42	Small	Secondary	Math	1	Non-digital	0,657
Bunch (2012)	96	Medium	High School	Science	6	Digital	−0,072
Can (2010)	44	Small	Secondary	Science	6	Non-digital	0,670
Canbay (2012)	52	Medium	Secondary	Math	other	Non-digital	0,740
Carr (2012)	104	Large	Primary	Math	9	Digital	0,386
Cengizhan and Koç (2017)	53	Medium	Secondary	Math	4	Non-digital	1,272
Chang, Evans, Kim, Norton, and Samur (2015)	266	Large	Secondary	Math	9	Digital	0,423
Chang and Hwang (2017)	65	Medium	Primary	Science	other	Digital	1,145
Chen, Yeh, and Chang (2016)	120	Large	Secondary	Science	8	Digital	0,148
Chen, Law, and Chen (2018)	129	Large	High School	Science	Unspcfd	Digital	0,205
Chen, Yang, Huang, and Fu (2019)	150	Large	University	Science	Unspcfd	Digital	0,284
Cheng and Su (2012)	64	Medium	University	Social Sciences	2	Digital	0,839
Cicchino (2015)	177	Large	Secondary	Social Sciences	Unspcfd	Non-digital	−0,216
Cooper (2018)	105	Large	Secondary	Math	9	Digital	−0,170
Çalışkan (2019)	120	Large	Primary	Math	Unspcfd	Non-digital	1,348
Çeker (2017)	38	Small	Primary	Science	7	Non-digital	2,044

(continued)

Appendix A. Continued.

Author and year	N	Sample size	Grade/Class	Course/Field	Weeks	Kinds of games	Effect size
Çoşkun (2012)	54	Medium	Secondary	Science	6	Non-digital	1,397
Çuha (2004)	67	Medium	Secondary	Math	other	Non-digital	0,511
Demir (2016)	54	Medium	Primary	Math	8	Non-digital	1,337
Demirel and Karakus	32	Small	Secondary	Foreign Langs.	11	Non-digital	1,652
Yilmaz (2019)—1							
Demirel and Karakus	41	Small	Secondary	Math	11	Non-digital	0,838
Yilmaz (2019)—2							
Demirkaya (2017)—1	50	Medium	Secondary	Social Sciences	8	Non-digital	0,339
Demirkaya (2017)—2	56	Medium	Secondary	Social Sciences	8	Non-digital	1,216
Demirkaya (2017)—3	56	Medium	Secondary	Social Sciences	8	Non-digital	0,159
Dinçer (2008)	58	Medium	Primary	Math	4	Non-digital	1,341
Donmuş (2012)	69	Medium	Secondary	Foreign Langs.	4	Digital	0,766
Dündar (2015)	34	Small	Secondary	Math	5	Digital	0,225
Edeş (2019)	30	Small	Secondary	Social Sciences	9	Non-digital	0,222
Eltem (2018)	25	Small	Secondary	Science	8	Non-digital	1,197
Erol (2019)	40	Small	Primary	Foreign Langs.	5	Non-digital	0,662
Ferguson (2014)	222	Large	High School	Math	3	Digital	—0,349
Forbess (2010)	33	Small	High School	Foreign Langs.	4	Digital	0,014
Gazeteci (2014)—1	45	Small	Secondary	Science	9	Non-digital	0,944
Gazeteci (2014)—2	51	Medium	Secondary	Science	9	Non-digital	0,883
Gazeteci (2014)—3	45	Small	Secondary	Science	9	Non-digital	0,931
Gençer (2016)	42	Small	Secondary	Science	4	Non-digital	0,996
Ghaemi and Ebrahimi (2015)	20	Small	Preschool	Foreign Langs.	3	Digital	0,662
Gökçen (2009)	40	Small	Secondary	Math	2	Non-digital	1,816
Göv (2019)	94	Medium	Secondary	Social Sciences	10	Non-digital	0,448
Gülay (2008)	54	Medium	High School	Social Sciences	12	Non-digital	0,190
Güler (2011)	50	Medium	Secondary	Science	2	Non-digital	0,624
Gülsoy (2013)	40	Small	Secondary	Foreign Langs.	5	Non-digital	0,970
Günel (2019)	96	Medium	Secondary	Foreign Langs.	4	Digital	1,386
Güngörmüş (2007)	50	Medium	University	Computer	2	Digital	0,615
Gürbüz, Çeker, and Toman (2017)	38	Small	Primary	Science	7	Non-digital	2,044
Gürer and Arslan (2017)	76	Medium	Primary	Social Sciences	3	Non-digital	0,258
Gürpınar (2017)	45	Small	Secondary	Science	9	Non-digital	1,720
Hava (2012)	34	Small	Primary	Math	2	Digital	0,962
Hieftje, Pendergrass, Kyriakides, Gilliam, and Fiellin (2017)	133	Large	Primary	Math	4	Digital	0,212
Hung, Hwang, Lee, and Su (2012)—1	65	Medium	Primary	Math	Unspcfd	Digital	—0,098
Hung et al. (2012)—2	65	Medium	Primary	Math	Unspcfd	Digital	0,357
Hwang et al. (2012)	50	Medium	Primary	Science	other	Digital	1,227
Hwang, Wu, Chen, and Tu (2016)	57	Medium	Primary	Science	2	Digital	0,589
Hwang, Yang, and Wang (2013)	56	Medium	Primary	Science	other	Digital	0,729
Işık (2016)	70	Medium	Primary	Foreign Langs.	5	Digital	1,343
Kamnardsiri, Hongsit, Khuwuthyakorn, and Wongta (2017)—1	31	Small	University	Foreign Langs.	4	Non-digital	1,626

(continued)

Appendix A. Continued.

Author and year	N	Sample size	Grade/Class	Course/Field	Weeks	Kinds of games	Effect size
Kamnardsiri et al. (2017)–2	31	Small	University	Foreign Langs.	4	Non-digital	0,786
Kamnardsiri et al. (2017)–3	31	Small	University	Foreign Langs.	4	Non-digital	1,527
Kamnardsiri et al. (2017)–4	31	Small	University	Foreign Langs.	4	Non-digital	1,828
Kamnardsiri et al. (2017)–5	31	Small	University	Foreign Langs.	4	Non-digital	1,783
Karabağ and Aydoğan (2015)–1	57	Medium	High School	Social Sciences	15	Non-digital	1,086
Karabağ and Aydoğan (2015)–2	57	Medium	High School	Social Sciences	15	Non-digital	0,935
Karabağ and Aydoğan (2015)–3	57	Medium	High School	Social Sciences	15	Non-digital	1,175
Kavasoglu (2010)–1	65	Medium	Secondary	Math	5	Non-digital	1,595
Kavasoglu (2010)–2	73	Medium	Secondary	Math	5	Non-digital	0,501
Kavasoglu (2010)–3	62	Medium	Secondary	Math	5	Non-digital	0,880
Kaya and Elgün (2015)	61	Medium	Primary	Science	4	Non-digital	1,305
Kayabaşı and Akbaş (2017)	45	Small	Primary	Science	3	Non-digital	–0,696
Kesgin (2019)	54	Medium	Secondary	Foreign Langs.	8	Non-digital	–0,220
Kılıç (2010)	46	Small	Primary	Math	2	Non-digital	0,654
Kim and Ke (2017)	132	Large	Primary	Math	other	Digital	1,155
Koç (2019)	34	Small	Secondary	Science	5	Digital	0,852
Koçoğlu and Koka (2018)	70	Medium	Secondary	Social Sciences	4	Digital	0,272
Korcu (2019)	30	Small	Secondary	Social Sciences	Unspcfd	Non-digital	1,666
Lin et al. (2013)	62	Medium	Secondary	Math	Unspcfd	Digital	0,453
Lin, Hwang, Fu, and Chen (2018)	68	Medium	University	Foreign Langs.	8	Digital	2,017
Little (2015)	34	Small	Secondary	Science	Unspcfd	Digital	–0,270
Martin (2018)	80	Medium	Primary	Math	7	Digital	–0,426
Masri and Najar (2014)	158	Large	Primary	Foreign Langs.	8	Non-digital	0,531
Meister (2018)	86	Medium	Secondary	Math	3	Digital	–0,296
Mercan and Aktaş (2018)	38	Small	Secondary	Math	6	Digital	0,692
Müller, Son, Nozawa, and Dashtestain (2018)	36	Small	University	Foreign Langs.	7	Digital	0,998
Noah (2019)	240	Large	Secondary	Math	3	Digital	1,624
Nur (2019)	31	Small	Secondary	Science	4	Non-digital	0,771
Obut (2005)	70	Medium	Secondary	Science	Unspcfd	Digital	1,633
Öz Aydın et al. (2014)	56	Medium	University	Science	Unspcfd	Non-digital	0,908
Özaslan (2006)	24	Small	Secondary	Foreign Langs.	8	Non-digital	1,630
Özbal (2009)	37	Small	Secondary	Foreign Langs.	4	Non-digital	0,684
Öztürk (2018)	184	Large	Primary	Foreign Langs.	6	Digital	1,462
Öztürk (2004)	47	Small	University	Foreign Langs.	5	Non-digital	–0,770
Pekşen (2019)	48	Small	Secondary	Foreign Langs.	3	Digital	0,687
Perini, Luglietti, Margoudi, Oliveira, and Taisch (2018)	62	Medium	University	Social Sciences	other	Digital	0,127

(continued)

Appendix A. Continued.

Author and year	N	Sample size	Grade/Class	Course/Field	Weeks	Kinds of games	Effect size
Qteefan (2012)	140	Large	Primary	Foreign Langs.	5	Digital	1,036
Salam, Hossain, and Rahman (2015)	86	Medium	Secondary	Math	3	Non-digital	2,043
Sayan (2015)	44	Small	Primary	Math	6	Digital	0,113
Sert (2009)	266	Large	High School	Computer	2	Digital	−0,188
Sevigen (2013)	45	Small	Preschool	Math	10	Non-digital	1,293
Siew, Geoffrey, and Lee (2016)	60	Medium	Secondary	Math	other	Digital	2,318
Songur (2006)	90	Medium	Secondary	Math	8	Non-digital	0,591
Soydan (2019)	66	Medium	Secondary	Math	6	Non-digital	0,586
Starkey (2013)	168	Large	Secondary	Math	9	Digital	0,178
Su and Cheng (2013)	68	Medium	Primary	Science	Unspcfd	Digital	0,741
Su (2008)	145	Large	University	Computer	10	Digital	0,154
Sung and Hwang (2013)	62	Medium	Primary	Science	other	Digital	0,892
Swearingen (2011)	220	Large	High School	Math	7	Digital	−0,004
Şahin (2016)	38	Small	Secondary	Math	7	Non-digital	0,012
Şahin and ArslanNamlı (2016)	20	Small	Secondary	Science	8	Digital	0,537
Şahin (2015)	53	Medium	Secondary	Science	3	Non-digital	−0,111
Şaşmaz Ören and Erduran (2004)	33	Small	Secondary	Science	3	Non-digital	1,912
Şenol (2007)	40	Small	Secondary	Foreign Langs.	4	Non-digital	0,893
Tarnig and Tsai (2010)	60	Medium	Secondary	Social Sciences	5	Digital	0,897
Tayfur (2019)	48	Small	Secondary	Science	10	Non-digital	2,371
Tokgöz (2017)	79	Medium	Secondary	Science	3	Non-digital	0,354
Torun (2011)	56	Medium	Primary	Social Sciences	4	Non-digital	−0,946
Tuna Gündoğdu (2019)–1	47	Small	Secondary	Foreign Langs.	7	Digital	1,431
Tuna Gündoğdu (2019)–2	45	Small	Secondary	Foreign Langs.	7	Digital	0,760
Tural (2005)	52	Medium	Primary	Math	5	Non-digital	0,716
Tural Sönmez (2012)	75	Medium	Secondary	Math	8	Digital	0,688
Turan, Köklükaya, and Yıldırım (2020)	31	Small	Secondary	Science	4	Non-digital	0,771
Tut (2018)	64	Medium	Primary	Science	7	Non-digital	0,222
Türkmen (2018)	50	Medium	Secondary	Math	14	Digital	0,269
Türkmen and Soybaş (2019)	50	Medium	Secondary	Math	14	Digital	0,028
Türkoğlu (2016)–1	44	Small	Preschool	Social Sciences	12	Non-digital	0,613
Türkoğlu (2016)–2	44	Small	Preschool	Social Sciences	12	Non-digital	0,656
Türkoğlu (2016)–3	44	Small	Preschool	Social Sciences	12	Non-digital	0,635
Türkoğlu (2016)–4	44	Small	Preschool	Social Sciences	12	Non-digital	1,081
Ural (2009)	133	Large	Primary	Science	2	Digital	0,232
Usta et al. (2018)–1	39	Small	Secondary	Math	4	Non-digital	1,154
Usta et al. (2018)–2	39	Small	Secondary	Math	4	Non-digital	0,694
Usta et al. (2018)–3	39	Small	Secondary	Math	4	Non-digital	0,912
Usta et al. (2018)–4	39	Small	Secondary	Math	4	Non-digital	0,703
Varzani (2013)–1	56	Medium	Primary	Math	Unspcfd	Digital	0,086

(continued)

Appendix A. Continued.

Author and year	N	Sample size	Grade/Class	Course/Field	Weeks	Kinds of games	Effect size
Varzani (2013)–2	56	Medium	Primary	Science	Unspcfd	Digital	0,073
Varzani (2013)–3	56	Medium	Primary	Social Sciences	Unspcfd	Digital	0,000
Varzani (2013)–4	56	Medium	Primary	Foreign Langs.	Unspcfd	Digital	0,000
Wale (2013)	371	Large	Secondary	Math	Unspcfd	Digital	–0,108
Wang, Chang, Hwang, and Chen (2018)	107	Large	Secondary	Math	other	Digital	0,343
Wardaszko and Podgórski (2017)	158	Large	University	Social Sciences	other	Digital	0,791
Watson-Huggins (2018)	61	Medium	Secondary	Math	Unspcfd	Digital	2,501
Wyk (2011)	110	Large	University	Social Sciences	12	Non-digital	0,246
Yağız (2007)	51	Medium	Secondary	Computer	2	Digital	–0,615
Yağmur Şahin (2013)	38	Small	Secondary	Foreign Langs.	4	Non-digital	1,554
Yang (2017)	50	Medium	Primary	Social Sciences	other	Digital	0,345
Yaşar (2019)	32	Small	High School	Science	8	Non-digital	2,427
Yavuzkan (2019)	100	Large	Secondary	Math	6	Digital	0,652
Yazıcıoğlu (2017)	52	Medium	Secondary	Science	4	Non-digital	0,601
Yeşilkaya (2013)	50	Medium	Secondary	Social Sciences	3	Non-digital	0,359
Yıldız (2019)	69	Medium	Secondary	Foreign Langs.	4	Non-digital	0,452
Yıldız, Simşek, and Ağdaş (2018)	54	Medium	Secondary	Science	4	Non-digital	0,837
Yılmaz (2006)	20	Small	Preschool	Math	12	Non-digital	1,249
Yien et al. (2011)	66	Medium	Primary	Science	4	Digital	0,624
Yiğit (2007)	47	Small	Primary	Math	2	Digital	0,038
Yumuşak (2014)	56	Medium	Primary	Math	6	Non-digital	1,180
Yüksel (2019)	68	Medium	Secondary	Foreign Langs.	6	Digital	0,636
Zangana (2018)	32	Small	Preschool	Foreign Langs.	6	Non-digital	2,547
Zengin (2019)	337	Large	Secondary	Foreign Langs.	4	Digital	0,243

Unspecified: Unspcfd.

Appendix B. Cohen kappa values of the themes in the study

Cognitive ...				Drawbacks ...				Suggestions ...			
K2				K2				K2			
K1	+	–	Σ	K1	+	–	Σ	K1	+	–	Σ
	28	2	30		30	3	33		26	2	28
	1	15	16		1	17	18		2	14	16
	Σ	29	46		Σ	31	51		Σ	28	44
Kappa: .858 $p < 0.05$				Kappa: .833 $p < 0.05$				Kappa: .804 $p < 0.05$			